

IZMIR UNIVERSITY OF ECONOMICS

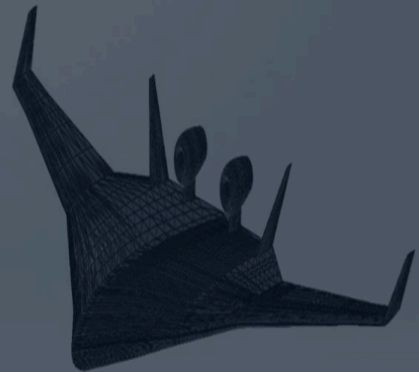
AE 405 Aircraft Design | Spring 2026

BWB-M1 "SPECTER"

Blended Wing Body Supersonic Strike Fighter

Conceptual Design Report

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Presentation Overview

- Design requirements (AE 405 Project)
- Aircraft geometry
- Methodology — Raymer 4-phase conceptual design
- Mission analysis & weight sizing
- Wing geometry & propulsion selection
- Aerodynamic analysis — drag buildup, L/D
- Performance results (turn rate ✓, ceiling ✓, max speed X)
- Weight breakdown & constraint diagram
- Design deficiencies & recommendations
- Requirements compliance summary

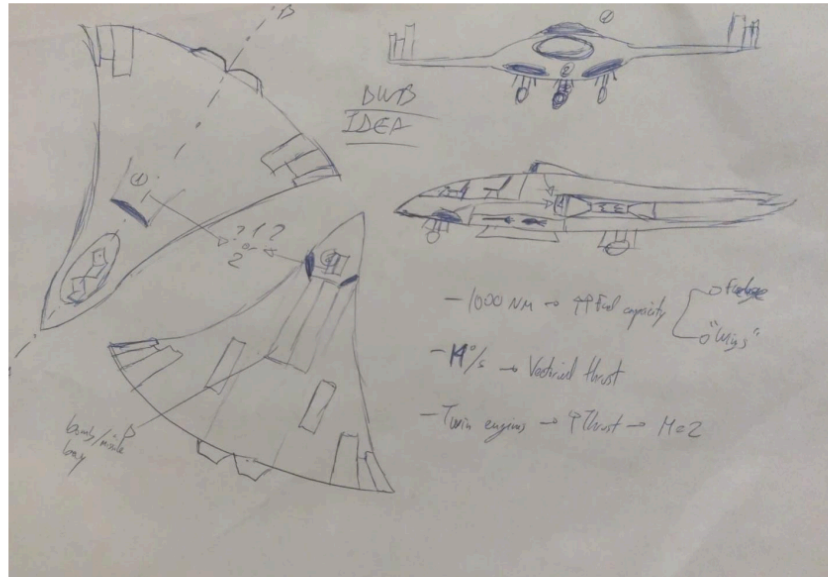
Design Requirements — AE 405 Project

Parameter	Requirement	Notes
Crew	2	Pilot + WSO
Mission Range	1,000 nm	2,000 nm with ext. tanks
Maximum Speed	Mach 2.0	Level flight
Sustained Turn Rate	≥ 14 deg/s	Combat maneuver
Service Ceiling	12,200 m (40,026 ft)	Max. operational altitude

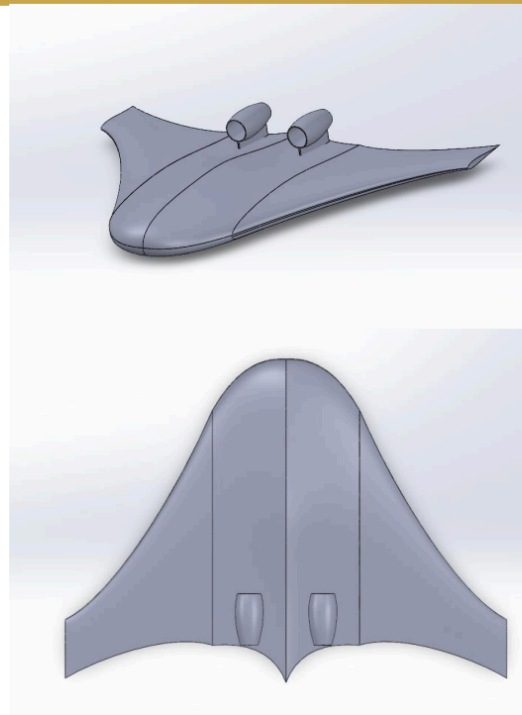
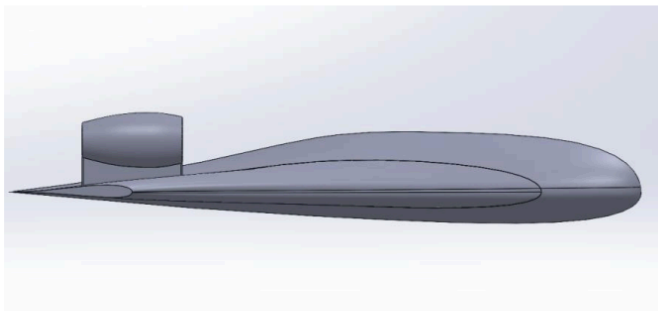
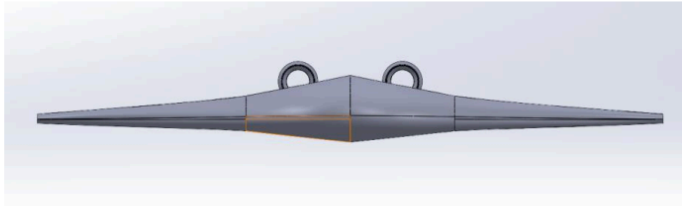
Aircraft Geometry

Blended Wing Body

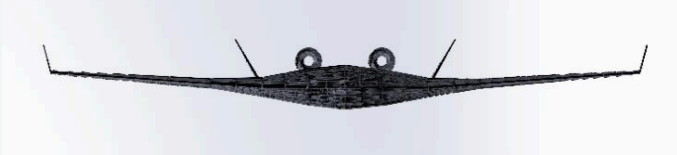
Paper sketch



Previous models



BWB-M1 Specter Geometry



Base Geometry — Boeing X-48

X-48 demonstrator (first flight 2007).

Wingspan $b = 37.5$ m, length $L = 18.5$ m. Downloaded 3-D model adapted for military use.

Modifications Applied

- ① Λ LE raised from 47° to 58° (supersonic LE sweep)
- ② Airfoil t/c reduced to 4% (outer panels)
- ③ Twin all-moving vertical stabilizers added
- ④ Engines repositioned to dorsal over-wing pods

Key Dimensions

Planform area $S = 384.4$ m²

Aspect ratio $A = 3.66$

MAC = 12.46 m

$\Lambda_c/4 = 54.1^\circ$

Methodology

Raymer — Aircraft Design: A Conceptual Approach, 6th ed.

Raymer 4-Phase Conceptual Design Process

01

Requirements Analysis

T/W, W/S, L/D, Mach number constraints from Table 1

02

Mission Analysis & Weight Sizing

Breguet fuel fractions → iterative W_0 sizing (Ch. 3)

03

Aero & Propulsion Sizing

Component drag buildup (CD0, K) → constraint carpet plot (Ch. 12)

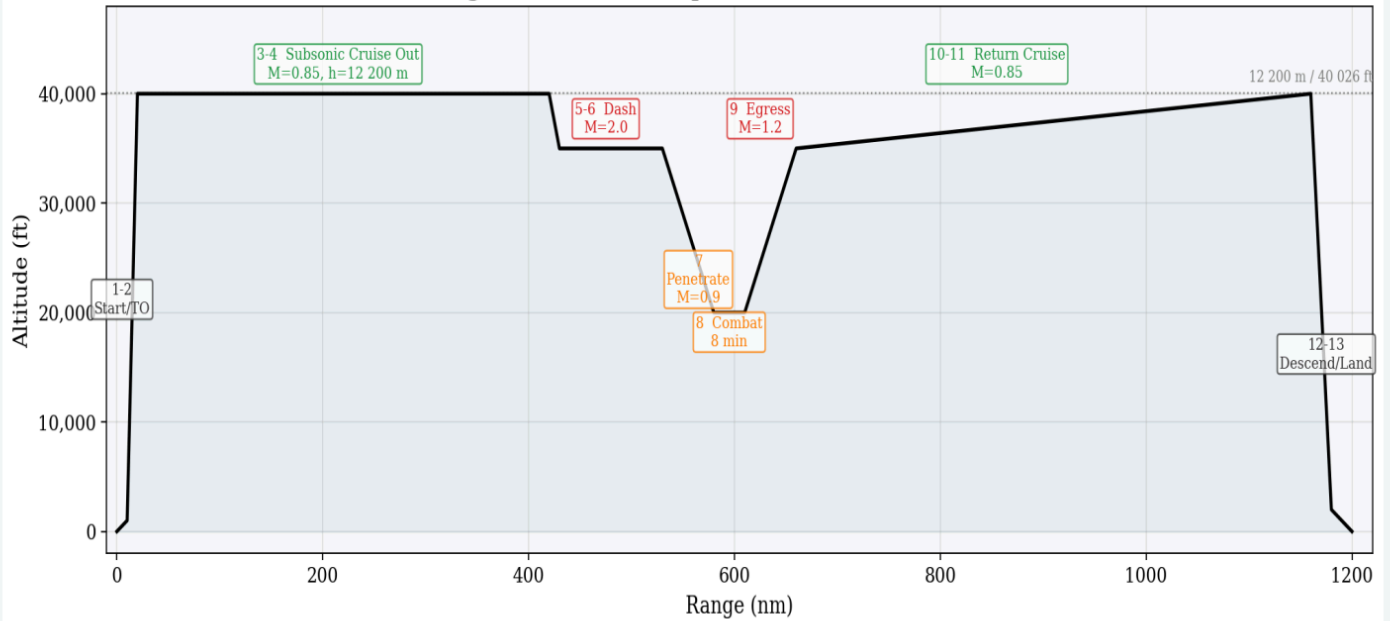
04

Performance Verification

Takeoff, climb, ceiling, max speed, turn rate — explicit pass/fail (Ch. 17, 19)

Mission Profile — 13-Segment Combat Strike Patrol

Figure 1 — BWB-M1 Specter: Combat Mission Profile



Mission Fuel Fractions

Segment	Description	W _i /W _{i-1}	Cumulative W _i /W ₀
4	Subsonic cruise out 400 nm (M 0.85)	0.951	0.905
5-6	Accel + Supersonic dash 100 nm (M 2.0)	0.960 / 0.967	0.841
7	Transonic penetration 50 nm (M 0.9)	0.983	0.827
8	Combat — 8 min, AB turns	0.930	0.769
10	Return cruise 500 nm (M 0.85)	0.940	0.708
13	Land	0.995	0.686

0.686

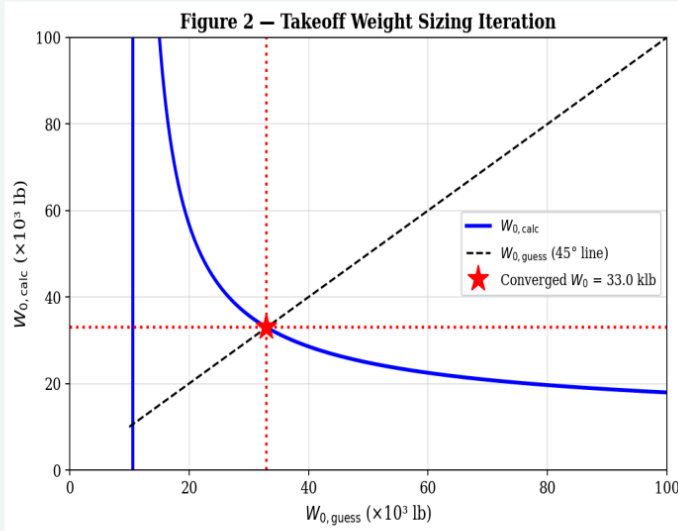
W_{ff} mission product

0.333

*W_f/W₀ fuel fraction
(+6% reserves)*

$$W_f/W_0 = 1.06 \times (1 - W_{ff}) = 0.333$$

Takeoff Weight Sizing — Raymer Iterative Method



Component	kg	fraction
W_0 Takeoff gross weight	14,963	1.000
W_e Empty weight	8,602	0.575
W_f Fuel weight	4,983	0.333
Crew + payload + weapons	1,379	0.092

38.9

W/S kg/m²
(F-22: 375 kg/m²)

14,963

W_0 kg
(32,988 lb)

⚠ Very low W/S — consequence of large X-48 BWB planform applied to 2-crew fighter mission

Propulsion & Aerodynamics

Engine selection, drag buildup, L/D analysis

Propulsion — 2 × GE F110-GE-132 Class Turbofan

1.97

*T/W at MTOW
(afterburner)*

1.18

*T/W at MTOW
(dry / MIL)*

289.2

*Total AB thrust
[kN]*

173.4

*Total dry thrust
[kN]*

Parameter	Value
Engine (per side)	GE F110-GE-132 class turbofan (AB)
AB thrust each	144.6 kN (32,500 lbf)
Dry thrust each	86.7 kN (19,500 lbf)
TSFC — subsonic cruise (M 0.85)	$\approx 0.82 \text{ hr}^{-1}$
TSFC — supersonic (M 2.0, AB)	$\approx 1.90 \text{ hr}^{-1}$
Installation	Dorsal over-wing pods, S-duct inlets
T/W comparison	F-22: 1.22 F-16C: 1.10 Specter: 1.97

Aerodynamic Analysis — Drag Buildup & L/D

0.00813

CD_0 subsonic
($M = 0.85$)

0.1061

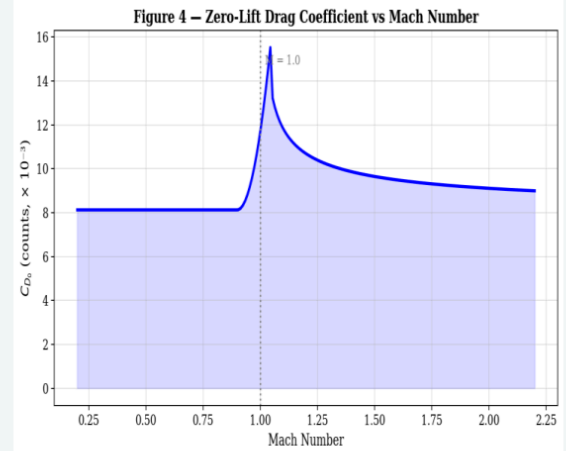
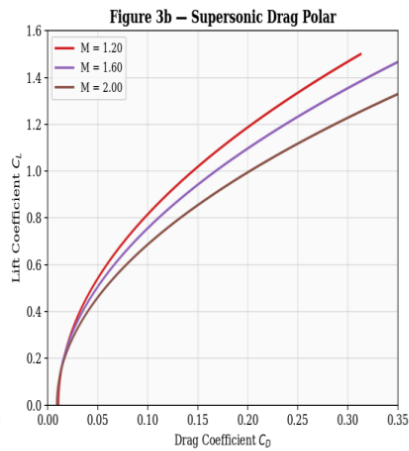
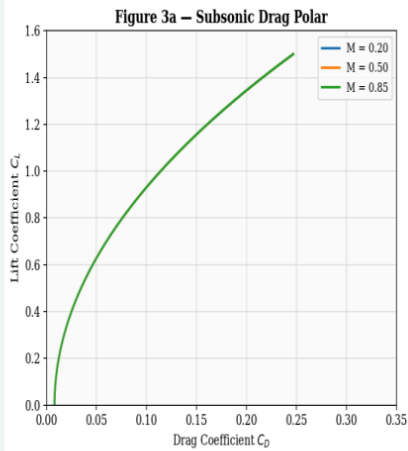
K induced drag
factor

17.0

$(L/D)_{max}$
 $M = 0.85$

0.277

CL_{opt}
at max L/D



Performance Analysis

Takeoff · Climb · Ceiling · Max Speed · Turn Rate

Performance: Takeoff & Climb

40 kt

Stall speed V_S
(MTOW, SL, $CL_{max}=1.5$)

44 kt

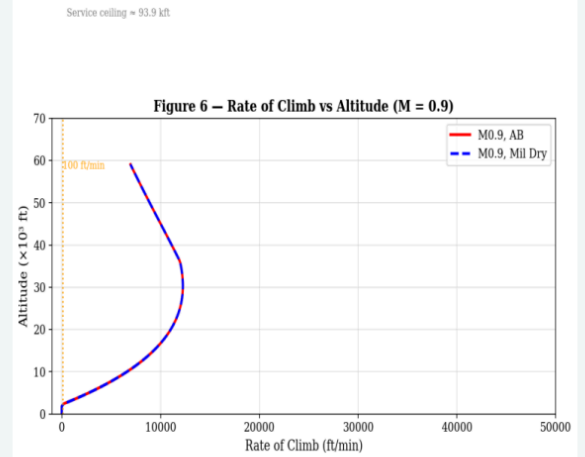
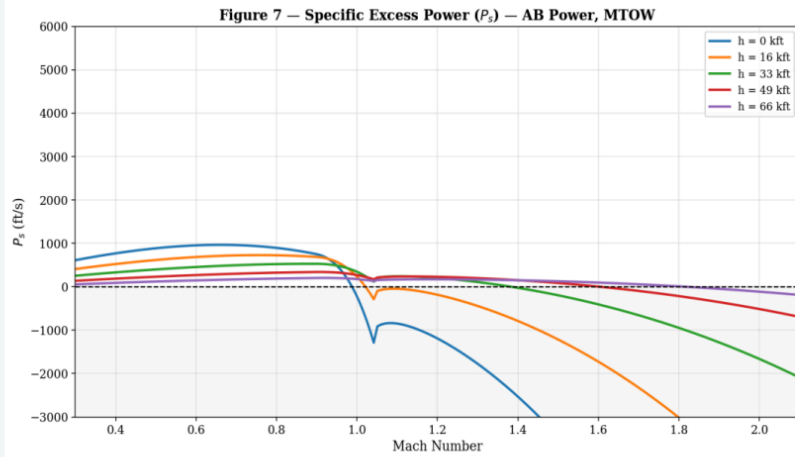
Takeoff speed V_{TO}
($1.10 \times V_S$)

31 m

Ground roll STO
(AB, dry ~200 m)

229 m/s

Max climb rate P_s
(SL, M0.9, AB)



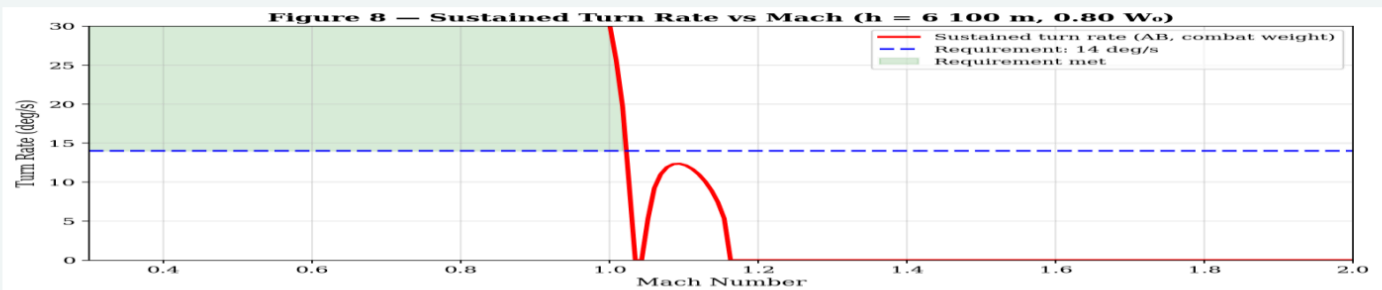
Performance: Sustained Turn Rate ✓ Requirement Met

14.5

deg/s sustained turn rate

✓ Requirement: ≥ 14 deg/s

- Evaluated: $M = 0.9$, $h = 6,100$ m, $W_c = 0.80 W_0$
- $q = 26,386$ N/m², $CL_{1g} = 0.0116$
- T/W available (AB) = 1.585
- Aerodynamic limit: $n_{aero} = 26.8$ g ($\gg$ struct. limit)
- Governing: structural limit $n_{struct} = 7.33$ g
- $\psi = n \times g / V = 7.33 \times 9.807 / 284.5 = 14.5$ deg/s



Performance: Service Ceiling ✓ Far Exceeded

>> 25,000 m

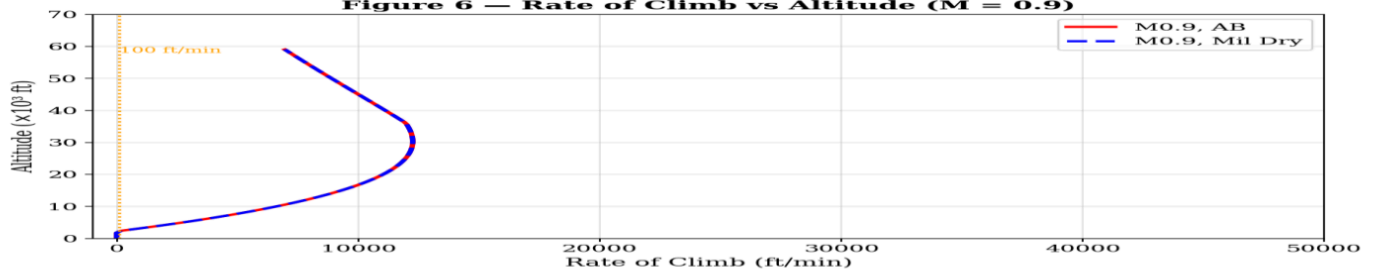
Theoretical ceiling at M 0.9, dry power

✓ Requirement: 12,200 m (40,026 ft)

- Installed dry T/W = 1.18 at MTOW — significantly above typical fighters
- Low parasitic drag ($CD_0 = 0.00813$) at subsonic cruise conditions
- Positive specific excess power maintained to very high altitudes
- SR-71 operates at 26,000 m; our design has similar T/W characteristics
- Requirement exceeded by a wide margin — no corrective action needed

Service ceiling ≈ 93.9 kft

Figure 6 — Rate of Climb vs Altitude (M = 0.9)



⚠ Maximum Speed — Requirement NOT MET



MACH 2.0 NOT ACHIEVABLE
in level flight at 12,200 m

Thrust available at M 2.0, h = 12,200 m (AB):

$T \approx 141 \text{ kN}$

Wave drag at M 2.0 ($CD_o \approx 0.040$, $S = 384 \text{ m}^2$, $q = 52,500 \text{ N/m}^2$):

$D \approx 808 \text{ kN}$

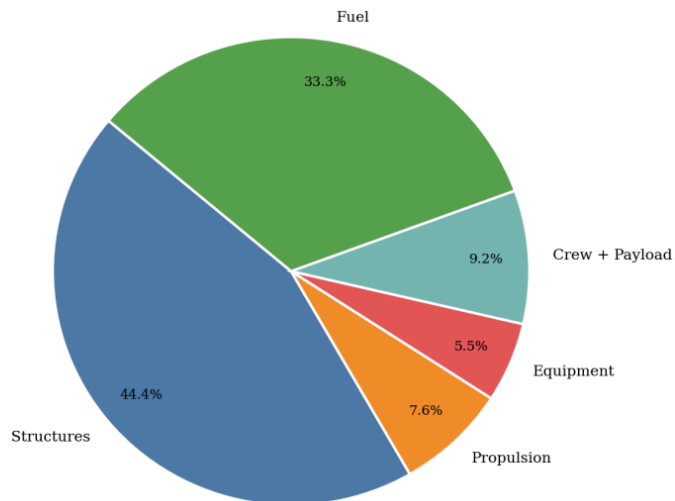
Thrust deficit: $T - D \approx -667 \text{ kN}$ (failure by factor ≈ 5)

Root cause:

- X-48 center body $t/c \approx 0.10\text{--}0.15$ — thick, designed for subsonic transport, not for Mach 2
- Very large planform $S = 384.4 \text{ m}^2$ amplifies wave drag: $D = CD_o \times q \times S$
- Wave drag scales as $(t/c)^2 / \sqrt{M^2-1}$ (Ackeret) — integrated over full span, total is prohibitive
- X-48 was never designed for supersonic flight — this is a configuration-level incompatibility

Weight Breakdown

Figure 10 — Takeoff Weight Breakdown
 $W_0 = 32988 \text{ lb (14963 kg)}$



Group	Key items	% of We
Structures	Wing, center body, fins, LG	~77%
Propulsion	Engines×2, nacelles, fuel sys.	~13%
Equipment	Avionics, controls, ECS, seats	~10%

8,602

Empty weight W_e [kg]
 $W_e/W_0 = 0.575$

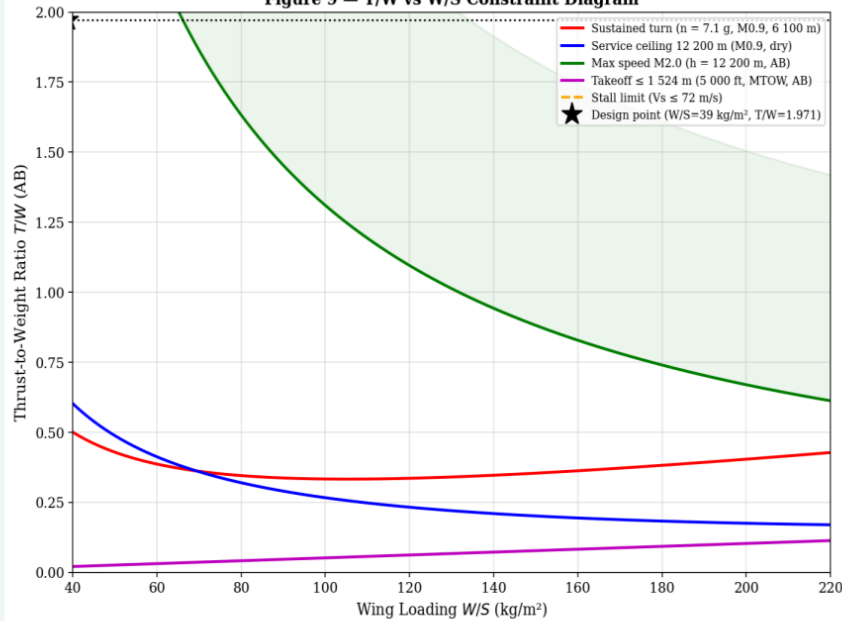
4,983

Fuel weight W_f [kg]
 $W_f/W_0 = 0.333$

Wing & center body dominate — characteristic of BWB where the central structure serves as fuselage, wing, and fuel/equipment bay simultaneously.

T/W vs W/S Constraint Diagram

Figure 9 — T/W vs W/S Constraint Diagram



Design Point

$W/S = 38.9$ kg/m²

$T/W = 1.97$ (AB)

Lies above ALL constraint lines — aircraft exceeds all constraints captured in the steady-level-flight model.

Constraints Shown

- ① Sustained turn ($n = 7.08$ g)
- ② Service ceiling 12,200 m
- ③ Max speed $M\,2.0$ (AB)
- ④ Takeoff $STO \leq 5,000$ ft

Note: The $M\,2.0$ constraint line does not capture the full BWB wave drag — the failure shown in Slide 18 requires the detailed drag model, not the steady-flight constraint.

Design Deficiencies & Recommended Corrections

X Mach 2.0 Not Met

X-48 center body wave drag exceeds AB thrust by $\times 5$ at $M=2$ level flight.

Fix: Area-rule center body, reduce $t/c < 6\%$, reduce S , or redefine req. as dive speed.

⚠ Low Wing Loading

$W/S = 38.9 \text{ kg/m}^2$ (vs F-22: 375) — X-48 transport geometry mismatched to 2-seat fighter mass.

Fix: Reduce planform area or increase mission payload substantially.

⚠ Structural Limit Dominance

Aerodynamic turn limit = 26.8 g $\gg$ structural limit = 7.33 g ($\times 3.7$ margin).

Fix: Re-evaluate structural design load factor; current design is aerodynamically over-capable.

Requirements Compliance Summary

Parameter	Computed	Requirement	Status
Mission Range (internal fuel)	1,000 nm	1,000 nm	✓ MET
Maximum Speed (level flight)	M << 2.0 (wave drag failure)	Mach 2.0	✗ NOT MET
Sustained Turn Rate (M 0.9, 6,100 m)	14.5 deg/s	≥ 14 deg/s	✓ MET
Service Ceiling	>> 12,200 m	12,200 m	✓ MET
Wing Loading W/S	38.9 kg/m ² (F-22: 375)	—	⚠ Anomalous
(L/D)max (M 0.85)	17.0	—	✓ High (BWB)
T/W at MTOW (AB)	1.97	—	— High

Conclusion

- Raymer methodology applied rigorously throughout — all four phases completed
- $W_o = 14,963 \text{ kg}$ | $W_e/W_o = 0.575$ | $W_f/W_o = 0.333$ — consistent convergence
- $(L/D)_{\text{max}} = 17.0$ and $CD_o = 0.00813$ — excellent subsonic efficiency from large BWB wing
- Requirements MET: mission range, sustained turn rate (14.5 deg/s), service ceiling (far exceeded)
- **Requirement NOT MET: Mach 2.0 max speed — X-48 center body wave drag exceeds thrust by $\times 5$**
- Non-compliance is inherent to applying a subsonic transport geometry to a supersonic mission
- Next steps: center-body area ruling, planform area reduction, or mission-profile redefinition

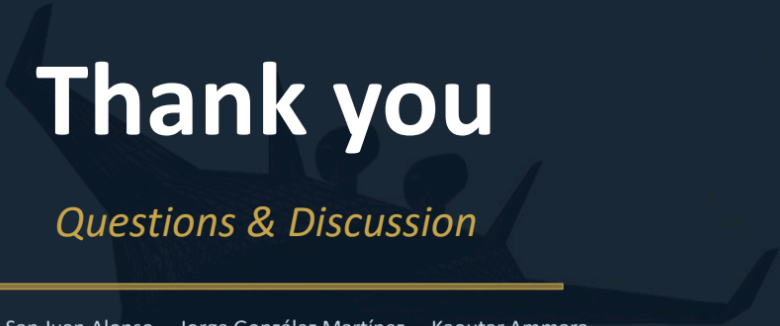
3/4

Requirements
met

1/4

Requirement
not met

*Design is internally consistent.
Non-compliances are genuine engineering
findings — not methodology errors.*



Thank you

Questions & Discussion

Sergio San Juan Alonso · Jorge González Martínez · Kaoutar Ammara
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